

KOSME CR22 M90 TECHNICAL DOSSIER

Electronic labeling-head control card: identity, repair evidence, component inventory, functional architecture, troubleshooting priorities, sourcing, and documentation gaps

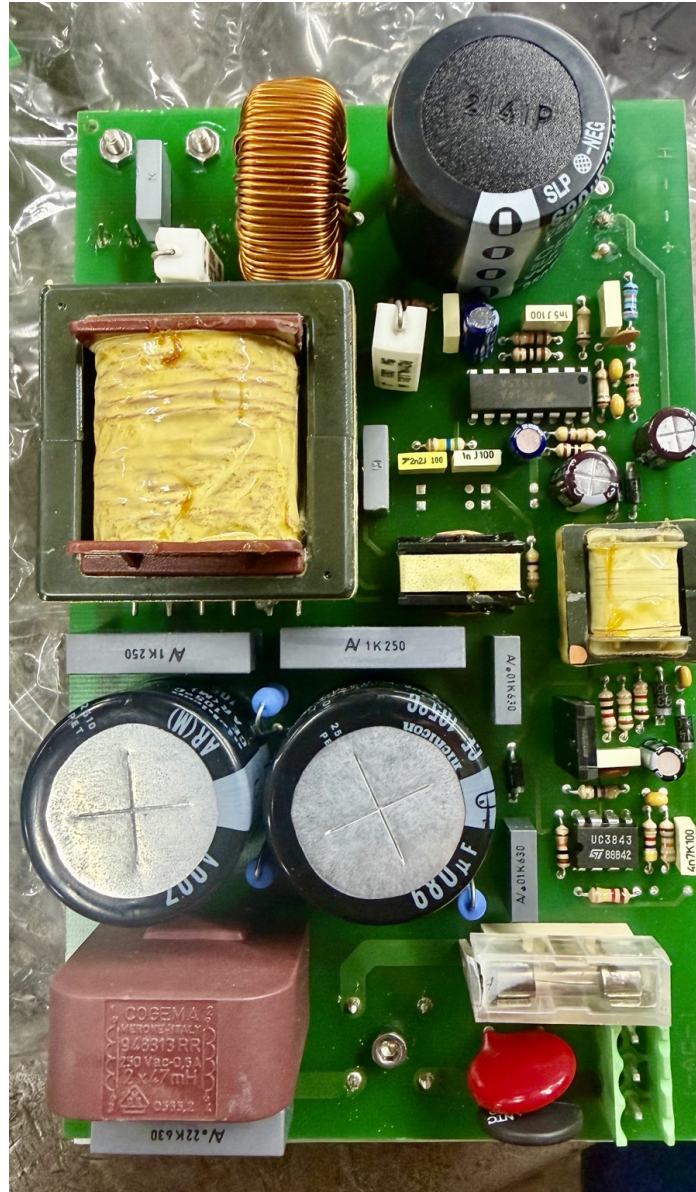


Figure 1. Internal power/control PCB photographed from the CR22 assembly. Original photograph supplied from the plant.

STATUS: The exact commercial assembly identity is confirmed online. One Radwell-replaced component is positively identified. No public factory schematic or connector pinout was located in indexed sources; circuit-level conclusions are therefore classified by confidence.

1. Equipment identity

Field	Current record	Evidence / confidence
Machine	Kosme TOP II AD CR30 rotary labeler	Plant record; high
Subsystem	Twenty electronic-cam / bottle-plate head assemblies	Plant observations; high
Controller	Elettronica Industriale CR22	Radwell report; high
Commercial listing name	Kosme Labeling CR22 M90 Labeling Head Control Board	Control Repair listing; high
Assembly architecture	CR22 with two attached boards	Control Repair listing; high
Affected positions	Heads/cards 15 and 20	Plant troubleshooting record; high
Observed alarm	Hardware error / network-card test hardware error	Plant troubleshooting record; high
Swap behavior	Fault followed the card when moved	Plant troubleshooting record; high
Addressing	Six-position binary DIP addressing: 1,2,4,8,16,32	Plant record; high

The strongest current diagnosis remains card-local failure because the fault followed the physical CR22 assembly during swapping. That behavior weighs against a fixed motor, cable, or carousel-position fault, although a complete loaded test is still required after repair.

The M90 term is present in current commercial listings for this exact Kosme board. It should be retained as an alternate model/search identifier even if it does not appear on every physical label.

Card address must remain separate from carousel position in the asset record. A board can be physically moved while retaining its DIP address, creating duplicate-address or wrong-address symptoms.

2. Radwell repair evidence



RADWELL
INTERNATIONAL, INC.


S3959900

20691157

REPAIR WORK REPORT

Technician:	Michael J. Pabst	Customer PO:	976220 // CREDIT CARDMA
Manufacturer:	ELETRONICA INDUSTRIALE	Customer ID:	1062750
Part:	CR22 (H)	Contact:	Doug Fischer
Serial Number:	N/A	Date:	8/20/2026

REPAIR DESCRIPTION

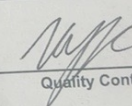
Removed and replaced components
Unit powers on with 120VAC applied. DC outputs measured good throughout terminal connectors. Unable to fully test unit due to limited assets.

Cleaned and Removed Corrosion
Component Tested
Functionally tested all Components
Power Tested under Load
Power Tested Unit
Reflowed Solder on PCB or Connections for reliability
Repaired Burned area of printed circuit board
Repaired Burned/Broken PCB traces
Repaired Cracked or Broken Solder Connection
Replaced Components
Unit Cleaned

*Returned to Plant
from Radwell 8-24-26
Have not installed &
Tested!*

PARTS USED

1 SLP681M200E3P3



 Quality Control Manager

Assuring Quality at Radwell International LLC
 Quality Assurance verifies that each unit has completed the applicable Radwell repair and/or inspection processes, that all required Radwell procedures have been followed, and that the unit has been correctly tagged and packaged. The unit is then protected using appropriate anti-static (ESD) packaging materials and prepared for shipment.

 Your unit will be packaged using suitable materials to help ensure safe transportation and returned with the applicable documentation. We recommend retaining the packaging materials for any future service requirements, as they can help protect your equipment during transit.

 If this unit was not repaired, a replacement unit may be available from stock. Please contact Radwell for availability.
Questions or Comments
 Please contact: The Returns Department at 800.332.4336, 609.288.9393, Ext 650 or email: returns@radwell.com

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Figure 2. Radwell repair work report dated August 20, 2026.

Report item	Recorded value
Technician	Michael J. Pabst
Manufacturer	Elettronica Industriale
Part	CR22; handwritten H
Customer PO	976220 // CREDIT CARDMA
Customer ID	1062750
Contact	Doug Fischer

Report item	Recorded value
Radwell identifiers	S3959900; 20691157
Work	Removed and replaced components; cleaned and removed corrosion
Bench result	Powered with 120 VAC; DC outputs measured good through terminal connectors
Test limitation	Unable to fully test due to limited assets
Part used	1 x SLP681M200E3P3
Plant note	Returned 8/24/26; not yet installed and tested

3. Positively identified replacement component

SLP681M200E3P3 is a high-voltage snap-in aluminum electrolytic capacitor. The part number on the Radwell report corresponds to the large SLP-series capacitor visible at the upper-right of the photographed board.

Parameter	Specification
Manufacturer family	Cornell Dubilier / Illinois Capacitor / Knowles SLP series
Capacitance	680 μ F
Rated voltage	200 VDC
Tolerance	$\pm 20\%$
Temperature rating	105°C
Mounting	Radial snap-in, 2 pins
Case	Approximately 30 mm diameter x 30 mm height
Lead spacing	10 mm
Ripple current	2.62 A (distributor specification)
ESR	Approximately 293 m Ω (distributor specification)
Rated endurance	3,000 hours at 105°C (distributor specification)

Interpretation

This is a DC-bus energy-storage/filter capacitor. At a 120 VAC bench input, a conventional full-wave rectifier produces approximately 170 VDC unloaded, which fits a 200 VDC capacitor rating. The visible toroidal choke and additional high-value capacitors indicate substantial filtering and/or an inverter power stage.

Radwell's replacement establishes that this capacitor was considered failed, degraded, damaged, or unsuitable. It does not establish that it was the only fault. Corrosion removal and the inability to run a full Kosme-system test leave the network, drive-output, feedback, and load behavior unverified.

Do not substitute only on capacitance and voltage. A replacement must also fit the snap-in footprint, polarity, diameter/height envelope, ripple-current demand, ESR, temperature rating, and service life.

4. Component-side inventory

The table inventories components visible in Figure 1. Exact reference designators cannot be assigned until the solder side and PCB silkscreen are photographed.

ID	Visible item	Marking / value	Confidence	Likely function
A	Large snap-in capacitor, upper right	SLP; Radwell PN SLP681M200E3P3	Confirmed	200 V DC-bus storage/filter
B	Toroidal copper-wound magnetic	No legible value	High: type; med: role	Series choke / current smoothing
C	Large EI laminated transformer	No legible PN	High: type	Isolation or low-frequency power conversion
D	Two large electrolytics, lower center	680 μ F visible; voltage obscured	Medium	Secondary or split-bus filtering
E	8-pin ST controller	UC3843; lot/date 86842	Confirmed	Current-mode PWM controller for switch-mode converter
F	Small ferrite transformer, center	No legible PN	High: type	High-frequency SMPS transformer / isolation
G	Small EI transformer, right	No legible PN	High: type	Auxiliary supply, feedback, or drive isolation
H	Brown COGEMA encapsulated magnetic	COGEMA 2; 9408313 RR1; 230 VAC-0.5 A visible	Partial	Custom magnetic or sensing component; topology required
I	Cartridge fuse and holder	Rating not visible	High: type	Overcurrent protection
J	Red disc	Marking not visible	Low	MOV or suppression capacitor; do not classify without marking
K	Black disc	Marking not visible	Low	NTC inrush limiter or MOV; do not classify without marking
L	16-pin DIP controller	Top code unreadable	Low	PWM/control/logic; exact part unknown
M	Cement power resistors	1K250; .01K630; 22K630 and other partial markings	Partial	Bleeder, limiting, snubber, sensing, or divider roles
N	Axial diodes and small electrolytics	Values partly obscured	High: classes	Rectification, clamp, startup, feedback, local filtering

Caution: the printed power-resistor text is not yet decoded into resistance, tolerance, and wattage with certainty. It must be read square-on and correlated to the copper traces before ordering replacements.

5. Functional architecture

The following block description is an engineering inference from visible topology, the 120 VAC bench test, the 200 V bus capacitor, the UC3843, and the CR22's application. It is not a recovered factory schematic.

Stage	Evidence	Probable operation	Verification needed
AC entry and protection	Fuse, disc components, power resistors	Fuse, surge suppression, inrush control, EMI attenuation	Terminal labels; trace map; disc markings
Rectification / high-voltage bus	200 V capacitor, choke, large magnetics	Converts nominal 120 VAC to a filtered high-voltage DC bus	Bridge/diode location on solder side; bus measurements
Auxiliary SMPS	UC3843, ferrite transformer, small electrolytics	Creates isolated low-voltage rails for logic, network, and gate drive	Transformer winding map and rail measurements
Drive power stage	Large capacitors and probable off-board/heatsink semiconductors	Supplies controlled energy to the bottle-plate motor/actuator	Other attached PCB and heatsink-device markings
Control / feedback	16-pin IC, resistor/capacitor network	Regulation, protection, current feedback, enable/fault behavior	IC top code and trace continuity
Network / addressing	Known CR22 card behavior and DIP address	Communicates head status/commands; reports hardware/network faults	Logic PCB, transceiver IC, connector pinout, bus waveform

Likely power path

120 VAC input → fuse/surge/inrush network → rectification → approximately 170 VDC bulk bus → filtered/controlled power stage → motor/actuator output. In parallel, the UC3843-based isolated converter likely derives low-voltage control rails. A loss or ripple fault on those rails can present as a network-card hardware error even when the communication wiring is intact.

The upper 200 V capacitor is therefore directly relevant to intermittent reset, undervoltage, excessive ripple, startup failure, or corrupted controller behavior. But only loaded ripple and rail measurements can prove that relationship.

6. Failure analysis for cards 15 and 20

Hypothesis	Supporting evidence	Contrary / missing evidence	Priority
Aged or failed DC-bus capacitor	Radwell replaced exact 680 μ F/200 V part; cards are legacy power electronics	No pre-repair ESR/capacitance report	Very high
Corrosion-caused leakage/open connection	Radwell explicitly cleaned and removed corrosion	Affected locations not documented	Very high
Auxiliary SMPS rail instability	UC3843 supply present; hardware/network faults can follow bad logic rail	No oscilloscope rail captures	High
Cracked solder joints / connector intermittence	Heavy magnetics and large capacitors mechanically stress joints	Solder side not photographed	High
Address conflict / incorrect DIP setting	Six-bit address determines identity; swapped cards can retain address	Fault followed card, favoring internal fault	Medium
Network transceiver failure	Alarm includes network-card test hardware error	Logic/network board not photographed	High
Motor/output-stage overload damage	Two affected heads were mechanically freer than others	Cards reportedly fault when moved; motor insulation/current data missing	Medium
Firmware/configuration mismatch	Replacement/cloning workflow and code-entry messages observed	Exact firmware/revision records absent	Medium

Commissioning gate

- Confirm every CR22 DIP address against physical head position before applying power.
- Record resistance and insulation-to-ground of the associated motor and cable with the card disconnected.
- Power the repaired card through the machine's normal protected supply; do not assume Radwell's unloaded 120 VAC result proves motor-drive operation.
- Measure DC-bus voltage and ripple, auxiliary logic rails, startup behavior, and thermal rise under load.
- Capture HMI alarms and network diagnostics before clearing or cloning data.
- Avoid cloning over the repaired card until its address, hardware revision, and original parameter contents are recorded.

7. Online sourcing and service intelligence

Source	What it contributes	Limitations
Control Repair	Exact CR22 M90 labeling-head control-board identity; two attached boards; repair and used purchase options; listed repair \$716 and used surcharge/purchase listing \$1,192	No public schematic; page access may be restricted
Krones.shop / Kosme catalog	Official route for machine-specific PCB and documentation lookup	Useful details require account and machine/serial association
Radwell	Actual plant repair record, part replaced, 120 VAC bench result, corrosion work	Could not fully function-test card
Cornell Dubilier / distributor records	Electrical/mechanical specifications for SLP681M200E3P3	Does not describe CR22 circuit role
Scribd Kosme manual listing	Evidence that a Kosme HMI/diagnostics/maintenance manual is publicly indexed	Not verified as the exact machine revision; no public CR22 schematic found

Recommended external request

Request the following from Krones/Kosme and both repair vendors using the exact identity: 'Kosme Labeling CR22 M90 labeling-head control board with two attached boards, Elettronica Industriale CR22.' Ask for the assembly drawing, electrical schematic, connector pinout, nominal DC rails, motor-output specification, DIP-address table, firmware/revision compatibility, test fixture procedure, known-failure list, and board-level parts list.

Control Repair contact published with the listing: 800-284-2420 and 716-836-2100. Their possession of a model-specific repair listing makes them a practical lead for reference photographs or a known-good comparison unit.

8. Exact evidence still required

These items prevent a defensible full schematic today. Acquiring them will convert the functional block model into a traceable board-level drawing.

Required evidence	Purpose	Minimum capture
Power PCB solder side	Trace every electrical connection and assign reference designators	Full-resolution perpendicular photo plus four overlapping close-ups
Second attached PCB, both sides	Identify network processor/transceiver, address switch, memory, firmware and connectors	Entire board plus IC markings
Third/other attached section if present	Complete the advertised two-attached-board assembly architecture	Both sides and interboard connectors
Heatsink / chassis devices	Identify MOSFETs, IGBTs, rectifiers, regulators, current sensors	Remove cover only when safely isolated; photograph every part code
Connector faces and silkscreen	Develop pinout and field wiring map	Straight-on image of each connector and wire labels
All capacitor sleeves	Build recapping BOM with dimensions and ratings	Rotate card; photograph complete markings
Known-good CR22	Compare component population, rework, jumper settings, revisions and resistance readings	Same photo sequence as repaired card
Krones electrical book pages	Recover official connector numbers, supply rails and network topology	Search by machine serial and CR22/M90 terms

Factory schematic status

No publicly indexed factory schematic, board-level BOM, or connector pinout for the CR22 M90 was found after searches using English and Italian terms, exact model phrases, the capacitor part number, Kosme TOP II identifiers, and repair-vendor records. This absence does not prove the document does not exist. The most likely repositories are the machine-specific Krones shop account, the original electrical drawing package, Kosme service archives, or repair-vendor internal files.

9. Asset Graph record

Entity	Proposed record
Asset type	Electronic cam / labeling-head motor control card
Canonical name	Kosme CR22 M90 Labeling Head Control Board
Manufacturer	Electronica Industriale; used by Kosme
Parent asset	Kosme TOP II AD CR30 labeler
Quantity / positions	20 head positions; confirm installed CR22 count and spares
Interfaces	120 VAC power confirmed for bench input; network and motor interfaces pending pinout
Addressing	Six-position binary DIP: 1,2,4,8,16,32
Known failures	Positions 15 and 20: hardware/network-card test hardware errors; fault followed card
Repair event	Radwell, 2026-08-20; returned 2026-08-24
Repair part	SLP681M200E3P3, 680 µF, 200 VDC, 105°C snap-in capacitor
Repair limitation	DC outputs measured good at terminals; no complete system/load test
Required relationships	Card → head position; card → motor; card → network node; card → repair event; card → documents/photos; card → spare/known-good comparison

Revision-control recommendation

Assign a unique plant asset ID to each physical CR22 card independent of its current head position. Record board revision, firmware, DIP address, serial/repair stickers, installation position, parameter backup, and every swap. This prevents the failure history of card 15 or 20 from being lost when cards move around the carousel.

10. Sources

1. Radwell International, Repair Work Report for Elettronica Industriale CR22, dated August 20, 2026. Plant-supplied image IMG_AFCF1694-7016-487E-BC03-F2105FBD0684.jpeg.
2. Plant-supplied CR22 internal PCB photograph, IMG_2840.jpeg.
3. Control Repair, 'Kosme Labeling CR22 M90 Labeling Head Control Board,' current indexed listing. <https://controlrepair.com/>
4. Control Repair, 'Kosme Labeling CR22 W/2 Attached Boards,' current indexed listing. <https://controlrepair.com/>
5. Krones.shop, Kosme spare-parts and printed-circuit-board catalog. <https://shop.krones.com/>
6. Mouser, SLP681M200E3P3 product record: 680 μ F, 200 V, $\pm 20\%$, 105°C snap-in aluminum electrolytic capacitor. <https://au.mouser.com/>
7. DigiKey, SLP681M200E3P3 product record and electrical/mechanical specifications. <https://www.digikey.com/en/products/detail/cornell-dubilier-knowles/SLP681M200E3P3/1882012>
8. RS, SLP681M200E3P3 product specifications including pin count, ripple current and lead pitch. <https://no.rs-online.com/>
9. Scribd, 'KOSME Labelling Machine Manual' indexed record. <https://www.scribd.com/>
10. Kosme, official company website. <https://www.kosme.com/>

Evidence classification

'Confirmed' means visible on the supplied artifact or explicitly stated by a directly relevant source. 'Probable' and 'inferred' describe engineering conclusions from component topology and application context; they are not substitutes for the factory schematic. Prices and availability are time-sensitive and must be rechecked before purchase.

End of dossier.